

Deep Learning for teens

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yolo

- YOLO does **real-time object detection** — it finds and labels objects in an image in one single pass.

```
dataset/  
├─ images/  
│   ├─ train/           # All .jpg or .png images for training  
│   └─ val/             # All .jpg or .png images for validation  
└─ labels/  
    ├─ train/           # YOLO formatted .txt annotation files for training  
    └─ val/             # YOLO formatted .txt annotation files for validation
```

yolo

- ▶ Vcop2 Image Dataset (annotated for Yolov8)

Yolo8 classes

0 person	21 bear	40 wine glass	60 dining table
1 bicycle	22 zebra	41 cup	61 toilet
2 car	23 giraffe	42 fork	62 TV
3 motorcycle	24 backpack	43 knife	63 laptop
4 airplane	25 umbrella	44 spoon	64 mouse
5 bus	26 handbag	45 bowl	65 remote
6 train	27 tie	46 banana	66 keyboard
7 truck	28 suitcase	47 apple	67 cell phone
8 boat	29 frisbee	48 sandwich	68 microwave
9 traffic light	30 skis	49 orange	69 oven
10 fire hydrant	31 snowboard	50 broccoli	70 toaster
11 stop sign	32 sports ball	51 carrot	71 sink
12 parking meter	33 kite	52 hot dog	72 refrigerator
13 bench	34 baseball bat	53 pizza	73 book
14 bird	35 baseball glove	54 donut	74 clock
15 cat	36 skateboard	55 cake	75 vase
16 dog	37 surfboard	56 chair	76 scissors
17 horse	38 tennis racket	57 couch	77 teddy bear
18 sheep	39 bottle	58 potted plant	78 hair drier
19 cow		59 bed	79 toothbrush
20 elephant			

- ▶ mAP50-95: 0.33070679075212683 # Overall detection + localization quality is low; boxes are not tight.
- ▶ mAP50: 0.6089962178523515 # Model detects objects fairly well at IoU 0.50; understands the class.
- ▶ Precision: 0.8800230121798416 # Predictions are mostly correct; very few false positives.
- ▶ Recall: 0.5294117647058824 # Model misses many real objects; high false negatives.

ML-flow

- ▶ MLflow logs metrics like accuracy, loss, MAE, MSE, so you can compare different training runs.

Assignment

- ▶ Pick another dataset. Try to the same steps just try to improve your result